

1 Tools of the Trade

Approximate numerical methods

1.1 Radioactive decay: time-step and convergence

Exercise 1. Using $N_0 = 10^{20}$, $\lambda = 1/2000 \text{ s}^{-1}$ and $\Delta t = 10 \text{ s}$, calculate N_1 using Forward Euler. Identify the corresponding numerical update.

Model answer. Forward Euler gives

$$N_1 = N_0 - \Delta t \lambda N_0 = 10^{20} \left(1 - \frac{10}{2000} \right) = 9.95 \times 10^{19}.$$

The corresponding Python statement is the update of the form `N_euler[i+1] = N_euler[i] - dt_euler*lam*N_euler[i]`. It is Equation 4 written directly as an assignment.

Exercise 2. Compare runs A and B. Has the analytical solution changed, or has the same exact function simply been sampled less frequently?

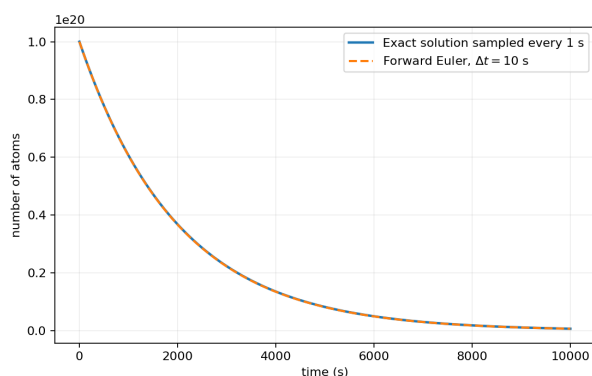
Model answer. The analytical solution has not changed. Run B uses a larger plotting interval for the exact solution, so the same exponential curve is represented by fewer sampled points. The sampled exact values still lie on the same decay curve, while the Forward Euler calculation is unchanged because `dt_euler` remains 10 s.

Exercise 3. Compare runs A, C and D. How does increasing the Forward Euler time-step affect the numerical solution? Why is a negative value of N unphysical?

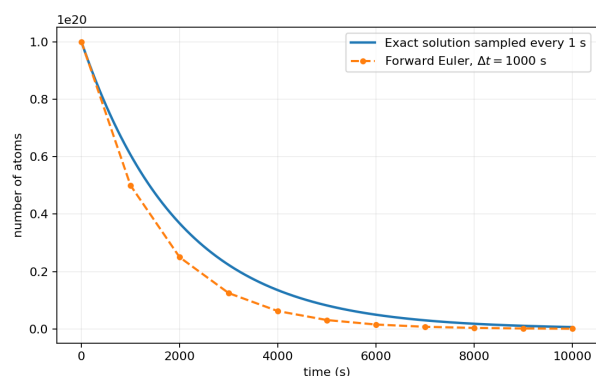
Model answer. As the numerical time-step grows, the Forward Euler solution departs progressively further from the exact exponential. With $\Delta t = 1000 \text{ s}$ the decay is much too rapid. With $\Delta t = 10000 \text{ s}$, $1 - \lambda \Delta t = 1 - 5 = -4$, so the very first Euler step changes the sign of N . A number of radioactive atoms cannot be negative; the negative value is therefore a numerical artefact caused by using a time-step that is far too large for this approximation.

Exercise 4. What should happen to the Forward Euler solution as `dt_euler` is reduced?

Model answer. It should approach the exact analytical solution. The truncation error introduced at each Forward Euler step becomes smaller as Δt decreases, so the numerical solution converges towards the continuous solution.



(a) Default: $\Delta t = 10 \text{ s}$.



(b) $\Delta t = 1000 \text{ s}$.

Figure 1: Increasing the Euler time-step increases the error. The 10 s calculation is almost indistinguishable from the exact solution, whereas the 1000 s calculation decays too rapidly.

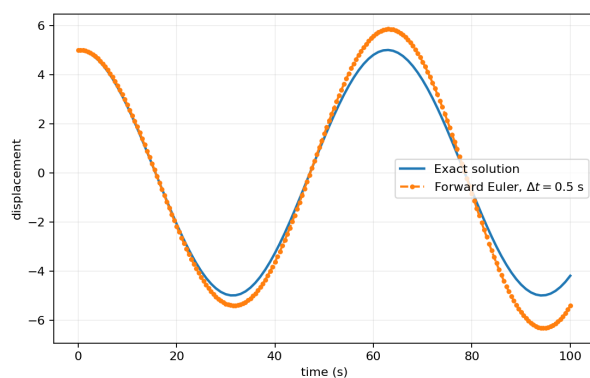
1.2 Second-order ODE: a harmonic oscillator

Exercise 5. The exact oscillator has constant amplitude. What happens to the amplitude of the Forward Euler solution with $\Delta t_{\text{euler}}=0.5$?

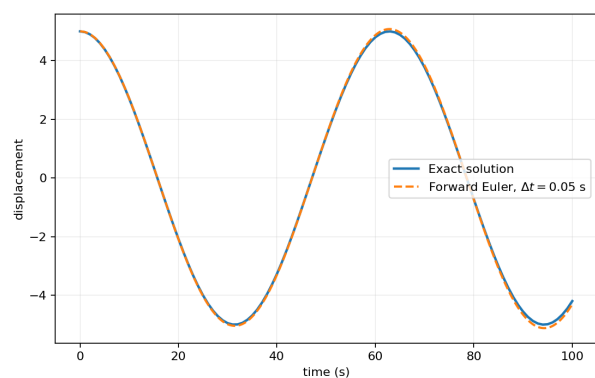
Model answer. The numerical amplitude grows with time. The Forward Euler method is adding numerical energy to this oscillatory system, so it does not preserve the constant-energy behaviour of the exact oscillator. There is also a growing phase error. The increasing amplitude is numerical, not physical.

Exercise 6. How does reducing the time-step to 0.05 s change the numerical solution?

Model answer. The numerical curve lies much closer to the exact solution: both the amplitude error and phase error are greatly reduced. This is another example of convergence as the numerical time-step is decreased.



(a) $\Delta t = 0.5$ s.



(b) $\Delta t = 0.05$ s.

Figure 2: Forward Euler is poorly behaved for the oscillator at the larger time-step, but the solution improves strongly when the time-step is reduced.

1.3 Transport equation: numerical diffusion and stability

Exercise 7. For the default $C = 0.5$ calculation, describe the numerical diffusion seen when the upwind solution is compared with the exact translated plume.

Model answer. The exact triangular plume keeps its original shape while it is translated. The upwind solution becomes progressively broader and its peak becomes lower. Material is artificially smeared into neighbouring grid cells even though the governing equation contains no physical diffusion. This is numerical diffusion produced by the first-order discretisation.

Exercise 8. What happens to the numerical diffusion when $C = 1$? Why is this value special for this simple problem?

Model answer. The numerical diffusion essentially disappears. With $C = v\Delta t/\Delta x = 1$, the plume moves exactly one grid cell during each time-step. For this one-dimensional constant-velocity upwind scheme, the update then reduces to an exact one-cell shift, so the grid values are translated without averaging between neighbouring cells.

Prediction. For $\Delta t=6$ s, calculate C and predict the behaviour.

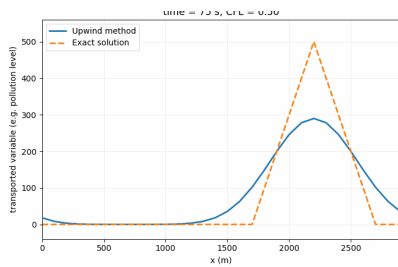
Expected prediction. With $\nu = 20 \text{ m s}^{-1}$ and $\Delta x = 100 \text{ m}$,

$$C = \frac{20 \times 6}{100} = 1.2.$$

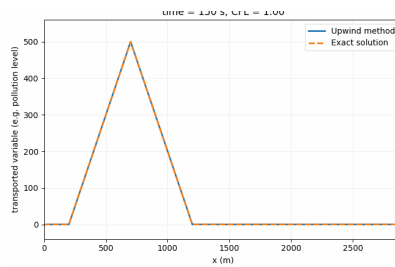
Because this exceeds the stability limit $C \leq 1$ for this scheme, the solution should become numerically unstable.

Exercise 9. What happens when $C = 1.2$?

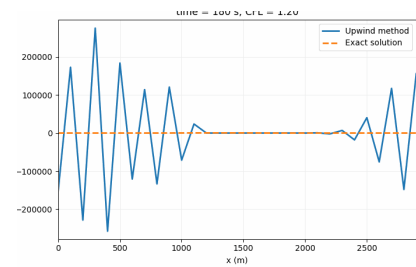
Model answer. Large oscillations develop and grow rapidly, including negative and very large positive values. The plume no longer resembles the physical translated solution. This is numerical instability caused by violation of the CFL condition, not a physical instability in the transport equation.



(a) $C = 0.5$: diffusive.



(b) $C = 1$: near-exact shift.



(c) $C = 1.2$: unstable.

Figure 3: Representative frames from the three upwind transport calculations.